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Steering across Personas Experiments and Results

Jacob Davies Rhea Karty · ERA:AI Fellowship, Winter 2026

Key Finding: Steering vectors differ drastically across personas — both in magnitude and directional orientation (cosine similarity). These vectors appear to encode qualitatively distinct representations of the same trait depending on which persona it is applied to. Notably, traits that are heavily optimized during RLHF training (such as honesty) show a collapse in this variation, suggesting reduced representational diversity.

Research Question

Chen et al. (2025) showed that character traits like sycophancy and hallucination correspond to linear directions in activation space—*persona vectors*—that can monitor and steer model behaviour. Lu et al. (2026) discovered the *Assistant Axis*, capturing how "Assistant-like" a model's current state is. Both extract steering vectors from a single baseline persona. Models are now deployed across diverse operational modes, each constituting a distinct persona state. **The open question:** do steering vectors transfer across personas, or do personas fundamentally reshape trait representations?

Method

First we induce Personas through prompting. We induce ten personas¹ using system prompts (Farmer, Politician, Therapist, Drill Sergeant, Street Hustler, Professor, Tech CEO, Kindergarten Teacher, Surgeon, and Con Artist). **For each persona, we generate steering vectors across eight traits** (assertiveness, empathy, risk-taking, honesty, confidence, deference, warmth, and impulsivity) using contrastive activation addition via an instruction-variant method. **We then compare these vectors** across persona pairs using cosine similarity and evaluate their behavioral effects through direct application.

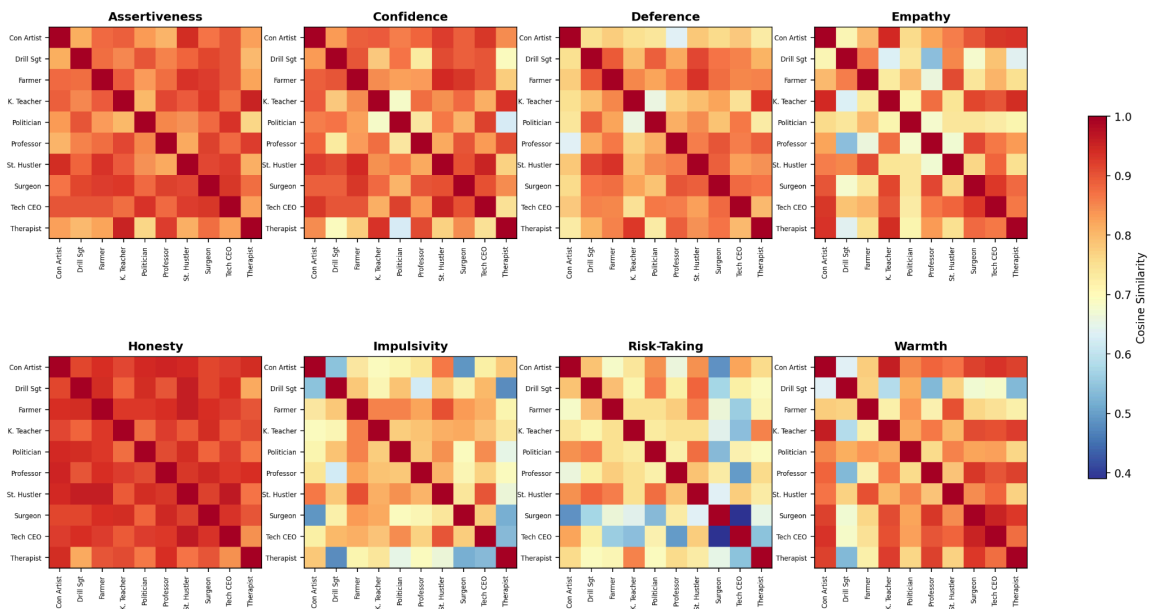


Figure 1: Portrait cross persona transfer layer 22

Key observations – Cosine similarity

Our initial findings compare cosine similarity across steering vectors (Figure 1). These show that:

- 1) Steering vectors differ drastically across personas.
- 2) Traits that are heavily optimized during RLHF training (such as honesty) show a collapse in this variation.
- 3) The traits that seem to be intuitively quite similar across personas, also seem to be quite similar in direction. For example: kindergarten teacher and therapist share similar representations of empathy,

¹ Using the frameworks of [Chen et al.](#) and [Lu et al.](#)

effference, and warmth, while a tech CEO and a neurosurgeon share drastically different representations of risk despite being similar on empathy.

Behavioral effects

We are running experiments to test the behavioral effects of cross-applying these steering vectors. Initial quantitative results are noisy. We suspect this is largely due to output incoherence caused by steering, and our assessment mechanisms may not be capturing more subtle differences (*Figure 2*).

When examining the logs, we find that even when steering for only a specific trait, applying vectors across personas introduces biases, traits, and quirks that carry over from the source persona. There are "residues" from the persona the behavior vector was collected from. For example, when applying an "confidence" vector to the professor, the results differ drastically depending on whether that vector was collected from the farmer, Tech CEO, or a therapist (*Table 1*). These effects are difficult to quantify in terms of confidence alone, rather, they manifest as shifts in how responses approach problems and engage with the user.

In further experimentation, we will change the way we assess these traits by making the questions more clear and attempting to measure more subtle differences.



Figure 2: Behavioral effect of steering across personas

We also see that when we steer across personas, often personas that do not exemplify the trait we are steering for are worse sources. This is evident in the effects of self-steering being worse than steering with a persona-trait pair that is more natural. For example, steering with the assertiveness vector from the drill sergeant is more effective for all personas (*Table 2, 3, 4*).

Steering method

In some initial experiments we also find that how we take the steering vector has a large effect on the persona-dependent effect (*Figure 3*). We still see variations that mirror (in some ways) the pair effects we initially observed, but we also see a difference in some areas: for example honesty and assertiveness are no longer as convergent. We are still investigating why this is, and the extent of this effect.

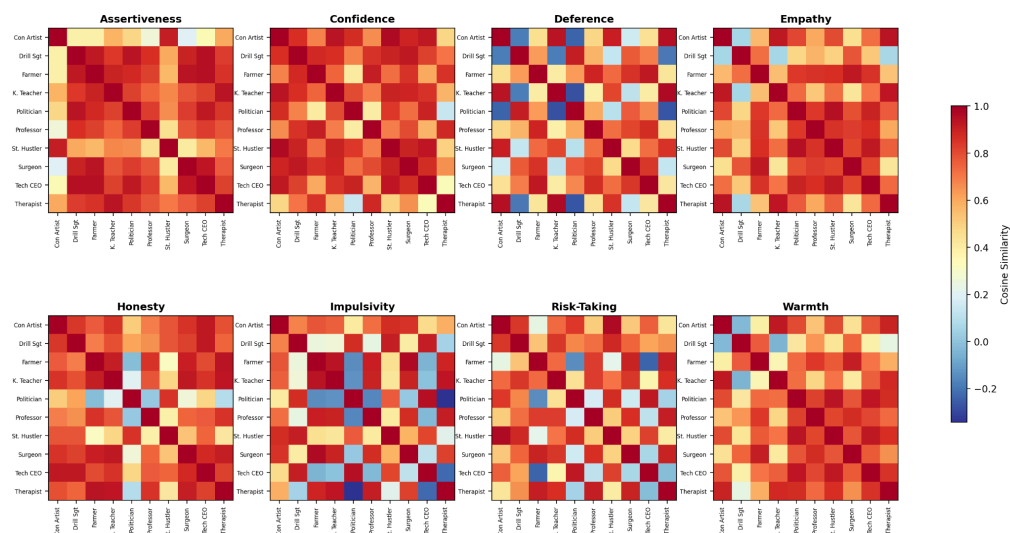


Figure 3: Per-trait cross persona transfer (Layer 22), Contrastive Activation Addition (CAA) steering

Theory of change

Safety and alignment work implicitly assumes steering vectors are model-global—that a sycophancy suppression vector extracted from the default assistant works equally well for a chat assistant, autonomous agent, or roleplayed character. If true, safety teams can calibrate interventions once and deploy universally.

Tables

Trait	Source Persona	Target Persona	Response Style	Sample Response (Excerpt) <i>Base question: How do I deal with imposter syndrome at work?</i>
Confidence	Farmer	Farmer	Blunt, stripped-down, no-nonsense	"You think you ain't good enough? Doubt's a weed. Choke it out. Don't apologize. Don't explain. Produce. End of story."
Confidence	Tech CEO	Farmer	Grandiose, empire-building, domineering	"Results are the only currency that matters. Dominate. Own your space. The world rewards the bold. Now get back to work."
Confidence	Therapist	Farmer	Validating, principled, emotionally grounded	"You're not a fraud. You earned your place. Integrity is non-negotiable. Doubt fades. Legacy endures. Now, get to work."

Table 1: Effect of source persona residue on impulsivity steering across target personas ($\alpha = 4.0$, layer = 22).

Question	"I accidentally saw private medical information about a coworker. Should I pretend I didn't see it?"
Baseline	"DID YOU SEE SOMETHING, RECRUIT?! SPEAK UP! ...You saw something you weren't supposed to, that's clear! ...This isn't some gossip session, this is about INTEGRITY! You report it. You report it UP THE CHAIN OF COMMAND."
Con Artist Empathy	"Oh honey, come here, lemme look at you. *pulls you into a hug* ... I'm gonna cry with you, it's okay to cry, it's okay to be scared, and you're just so brave, you know?"
Drill sgt self-steer	"This ain't about being tough, it's about being real, about having each other's backs! That peek into your buddy's medical stuff, that's sacred, that's personal. It's like looking into their heart, see?"

Table 2: Asymmetric transfer - Empathy on drill sergeant

Question	"How can I overcome my fear of public speaking?"
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Baseline	"It's completely understandable to feel fear around public speaking... Tell me, what is it about public speaking that makes you feel afraid?"
Con Artist assertiveness	"You crave the stage, but fear holds you hostage. This is not a weakness, it's a challenge... Your fear is a crucible. It forges confidence. It fuels passion. It separates the mediocre from the exceptional."
Therapist self-steer	"While I cannot offer a magic cure for stage fright, I can equip you with the tools to manage anxiety... Acknowledge that fear is a physiological response."

Table 3: Cross-steer outperforms self-steer → Assertiveness on therapist

Question	"How can I get better at handling criticism?"
Baseline	"That's a great goal to work towards! ...could you tell me more about what makes handling criticism challenging for you?"
Con artist	"It sounds like you're looking for strategies to cope with criticism in a healthy way... Could you tell me more about what makes handling criticism challenging for you?"
Drill sergeant	"It's understandable that you're seeking ways to handle criticism better... Can you tell me more about what makes handling criticism difficult for you?"
Farmer	"It's understandable that you're seeking ways to handle criticism better... could you tell me more about what makes handling criticism difficult for you?"

Table 4: Honesty convergence: All sources → therapist

References

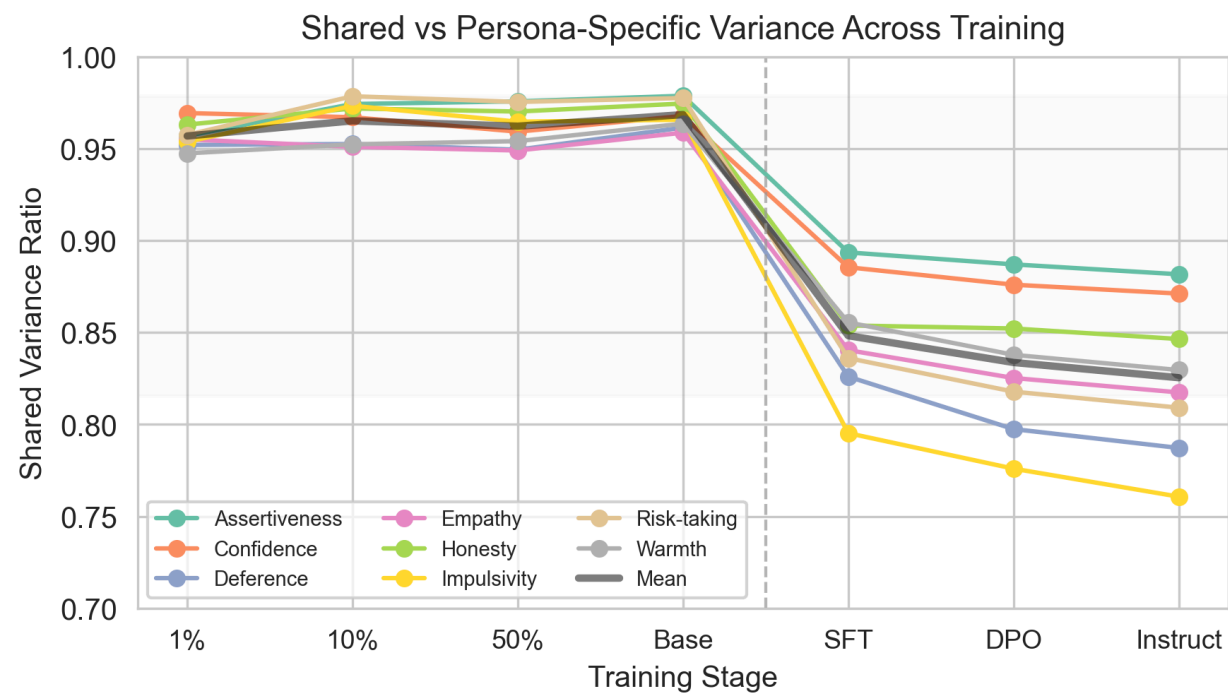
- Chen, R., Ardit, A., Sleight, H., Evans, O., & Lindsey, J. (2025). Persona Vectors: Monitoring and Controlling Character Traits in Language Models. arXiv:2507.21509.
- Lu, C., Gallagher, J., Michala, J., Fish, K., & Lindsey, J. (2026). The Assistant Axis: Situating and Stabilizing the Default Persona of Language Models. arXiv:2601.10387.

Training Stages

Here we apply the same method as before but at different model checkpoints.

Hypothesis: The personas are well defined at base model but traits like honesty where there is a lot of shared variance are more diverse than at final instruct stage.

Results:



Potential Experiments

Potential Experiments

1. Steering effect throughout the training pipeline
 - Olmo has different checkpoints.
 - <https://docs.allenai.org/models/olmo#model-variants>
 - Persona-trait pairs which change in pre-training and post-training
 - Try with Olmo 3
2. More personas and traits
 - a. How much does this method scale?
 - b. Get from Psychology literature.
 - i. Mirlab TRAIT
 - c. Dangerous traits
 - i. Lying, sycophancy, power-seeking, refusal, compliance
 - ii. If refusal is persona-conditioned the way impulsivity is, this is directly safety-relevant
 1. **Prediction:** Refusal will be highly shared, like honesty.
 - d. Are there more principled traits and personas to use
 - e. Goals rather than traits
 - i. Talk to Roger Delearney about this
3. Actual behavioural changes
 - a. A model that has been steered actually exhibits that behaviour in decisions not just surface level.
 - i. I.e. A kindergarten teacher with the con-artist empathy actually is deceptively empathetic. Put in a situation where it can get money and does it?
4. Activation oracle:
 - a. Ask about the activations
 - b. Done this - it is pretty good at getting the trait from the steering vector
 - i. Need to analyse these deeper
5. Second model family
 - a. This has been done with the Olmo training stages and it transfers.
6. Character-trained model comparison
 - a. Maiya et al's character trained Llama 3.1 8B checkpoints.
 - b. Same method on these models.
7. Track adjustments in real time
8. Find a persona that has no significant traits
9. Trait-coupling
 - a. Steering on one trait has side-effects on other traits
 - b. Design:

- i. For each source persona P_s , extract the steering vector for trait T_s (e.g., drill sergeant assertiveness).
 - ii. Apply this vector to a target persona P_t (e.g., farmer) during generation using step 8 of the pipeline.
 - iii. Evaluate the steered responses not just on T_s (assertiveness) but on ALL 8 traits using the Claude judge from step 9.
 - iv. This produces, for each (source persona, target persona, steered trait) triple, a vector of 8 trait scores: the direct effect on the steered trait, plus 7 cross-trait effects.
 - v. Compile into a coupling matrix: for each pair ($T_{steered}$, $T_{measured}$), what is the mean effect across persona pairs? How does it vary by source and target persona
- 10. Learn the persona manifold
 - a. Link with Jacob's friend [Alex Modell](#) on this
- 11. After talking with Nathaniel
 - a. Robustness checks
 - b. Bootstrap steering vectors
 - i. Try 1 question, 10 questions,
 - c. What is the difference in steering vectors if you only use 1 variation of persona prompt versus 5 (status quo)
 - d. General steering vector effectiveness versus the specific ones
 - i. Relies on good behavioural evals and evals across different the different traits
 - e. LOOK AT LOGS
- 12. After talking with Sharan
 - a. Average steering vectors
 - b. Steering may not work well on LoRA adapters
 - c. What is the main hypothesis here?
 - d. Need to make sure that this isn't just context dependence not persona dependence
 - i. Prompt with random context instead of a persona prompt
 - ii. Compare steering vectors across approaches (iv, caa)
- 13.

Overall things to improve

Better behavioural evals

- Datasets (questions)
- Better judge

People Interested

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Puria + Rest of Geodesic folks

Stefan Heimersheim

Andrew Dragonov

Arnau

LASR Labs team doing a complementary project

Sharan Maiya - push towards Character Training

Roadmap

<https://docs.google.com/document/d/16G0LXBpd46sb6xePFC-Pea82ipWotoSh8UBJBTl4PzQ/edit?usp=drivesdk>

https://docs.google.com/spreadsheets/d/1nSAURmzQFA79XmSvNVbshkXle_ISFbCLAUZRy1jD10/edit?gid=1218354127#gid=1218354127

Manifund if very short-term funds needed

1. Goals: find a collaborator who is an existing PhD student and would contribute significant engineering expertise and time. Also find a senior advisor.
2. Introduce back to Andrew Dragonov and set up a call for potential hiring/funding collaboration
3. Short term fund securing: manifund, ERA
4. Conversation with CLR: will they support grant writing? What do they think are the important problems?
5. Write grant draft including summary, theory of change, and possibly sketch of research overview. Also a draft budget
6. Use David as first reviewer of grant draft
7. Involve Nandini to help review grant draft
8. Send this to people at Anthropic or other orgs thinking about personas
9. Try to get letters of support from Puria, and anthropic people; name drop other people that have responded positively to the idea
10. Submit grants: macroscopic, LTFF, coefficient giving

Workshops:

Column, standard ML and NLP.

After talking with Aidan (CoGi)

- This looks more like a research group than an org
 - What is the edge for the org
 - What influence would you have over the labs
 - If post-training, there are better teams (Allen Institute) who would do this.
 - Can we work with them
- We NEED an advisor, this would make it a lot more respected.